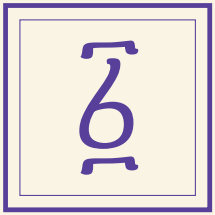
Finding the way, step by step

Weekend 1: Large language models

Carlos Cotrini
AI in Industry, ETH Zurich
Weekend 1, October 2026 𓂀𓋹𓂀𓆑

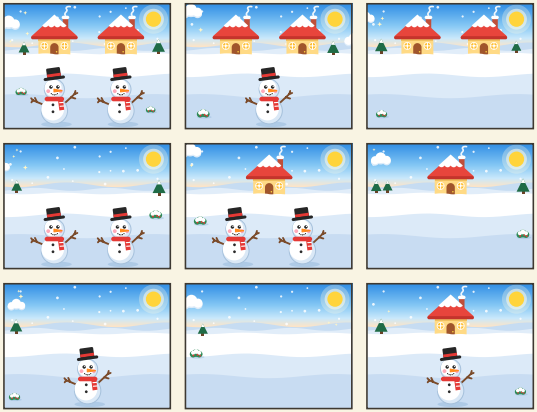


Section 1

The town

Mara carries letters across the town

Nine places. An invented town.



Mara carries letters across the town

Nine places. An invented town.

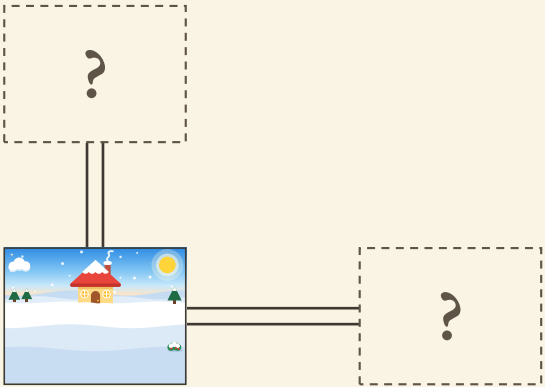
She starts at the meadow.



Every place has two roads

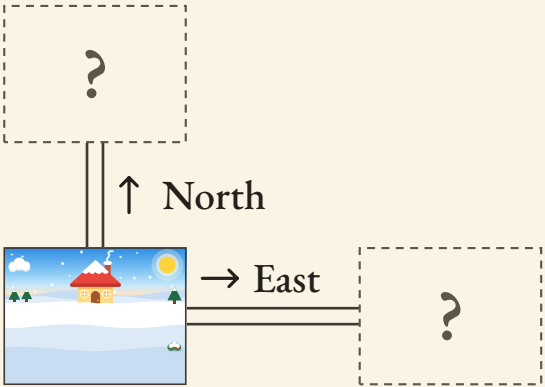


Every place has two roads



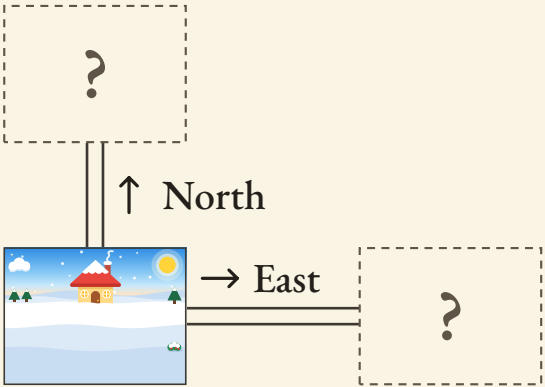
Where the roads lead, she finds out by walking.

Every place has two roads



Where the roads lead, she finds out by walking.

Every place has two roads



Where the roads lead, she finds out by walking.

At the edge of the town, one road is missing.

Each letter names a place



To

Each letter names a place



She starts at the meadow



Each letter names a place



She starts at the meadow



↑ North	→ East	✉ Deliver

Three things she can do.

Each letter names a place



She starts at the meadow



A coin if she delivers it there.

1

↑ North	→ East	Deliver

Three things she can do.

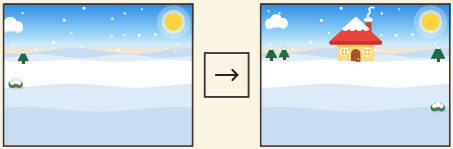
At first she can only guess





At first she can only guess





At first she can only guess





At first she can only guess





At first she can only guess



A guess pays about once in twelve letters, on average.



0
Paid nothing

Your turn: which road first?

Here



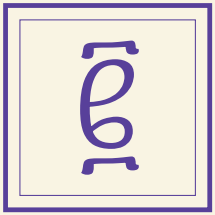
The letter



Which road first?

↑ North

→ East



Section 2

The clue

After many letters, a pattern

Letter to	Roads	In the picture
	→	
	→ →	
	↑	
	↑ ↑	

Letter to	Roads	In the picture
	↑ →	
	↑ → →	
	↑ ↑ →	
	↑ → → ↑	

After many letters, a pattern

Letter to	Roads	In the picture
	→	1
	→ →	2
	↑	0
	↑ ↑	0

Letter to	Roads	In the picture
	↑ →	1
	↑ → →	2
	↑ ↑ →	1
	↑ → → ↑	2

Every East road adds a house.

After many letters, a pattern

Letter to	Roads	In the picture	
	→	1	0
	→ →	2	0
	↑	0	1
	↑ ↑	0	2

Every East road adds a house.

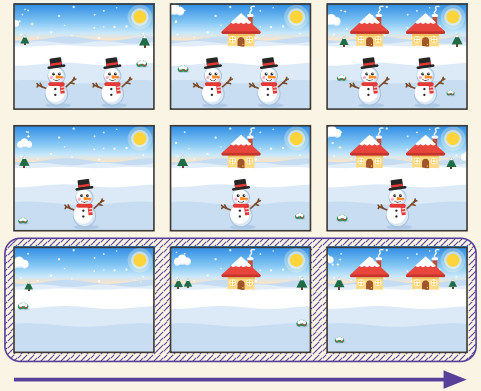
Letter to	Roads	In the picture	
	↑ →	1	1
	↑ → →	2	1
	↑ ↑ →	1	2
	↑ → → ↑	2	2

Every North road adds a snowman.

East adds a house, north adds a snowman



East adds a house, north adds a snowman



East: one more house at every road.

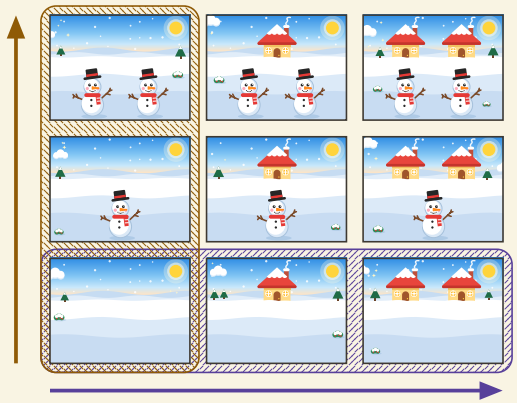
East adds a house, north adds a snowman



East: one more house at every road.

North: one more snowman at every road.

East adds a house, north adds a snowman



East: one more house at every road.

North: one more snowman at every road.

Nobody wrote this on the map. Mara had to notice it.

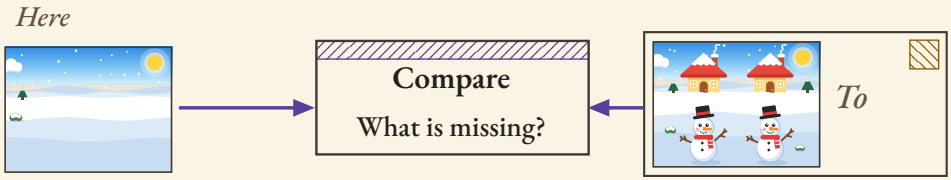
The rule: what is missing?

Here

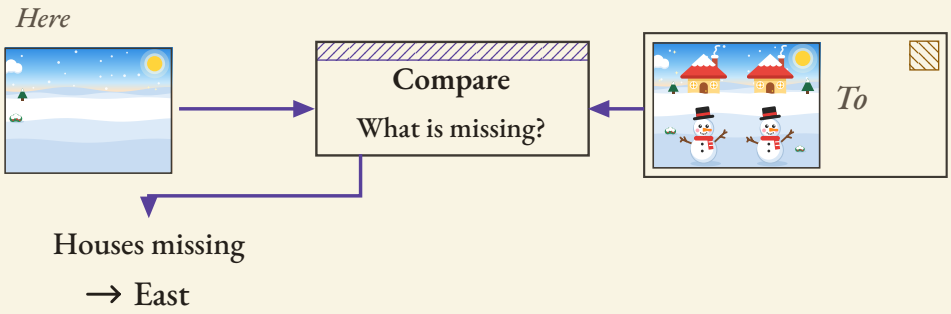




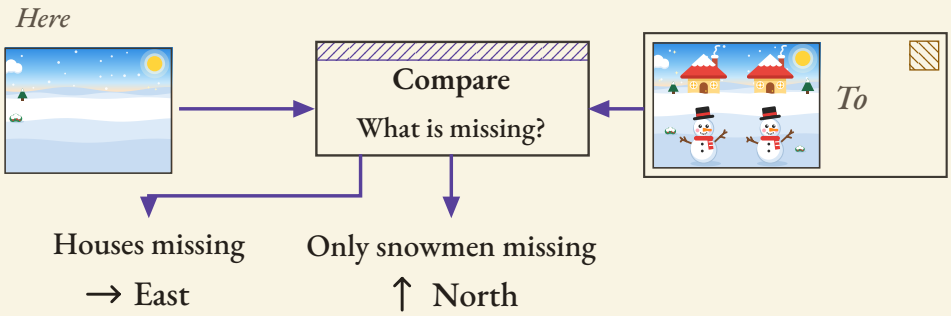
The rule: what is missing?



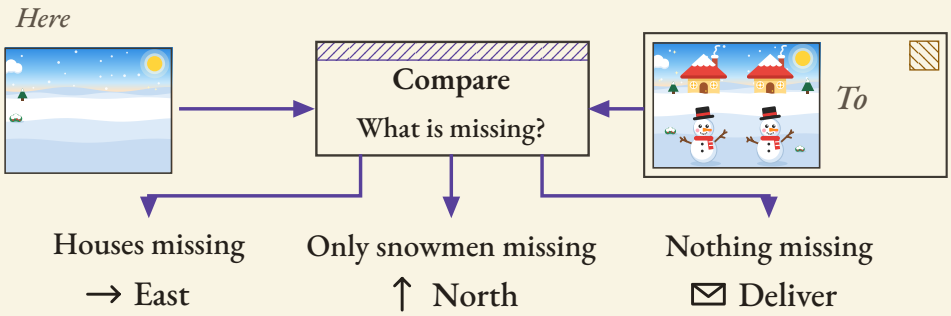
The rule: what is missing?



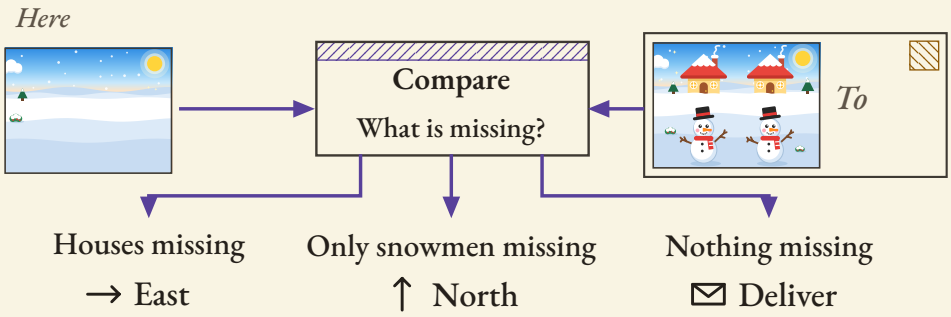
The rule: what is missing?



The rule: what is missing?



The rule: what is missing?



The sun, the clouds and the trees do not matter.

The rule finds the corner

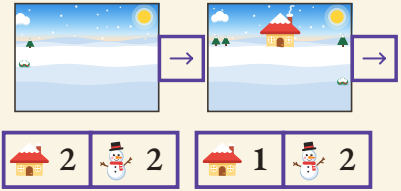






The rule finds the corner





What is missing decides the road.

The rule finds the corner



What is missing decides the road.

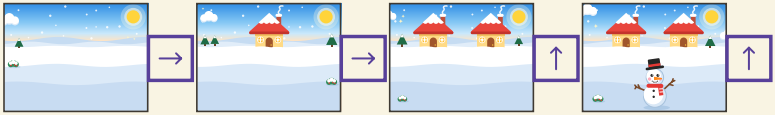


2 2 | 1 2 | 0 2

The rule finds the corner



What is missing decides the road.

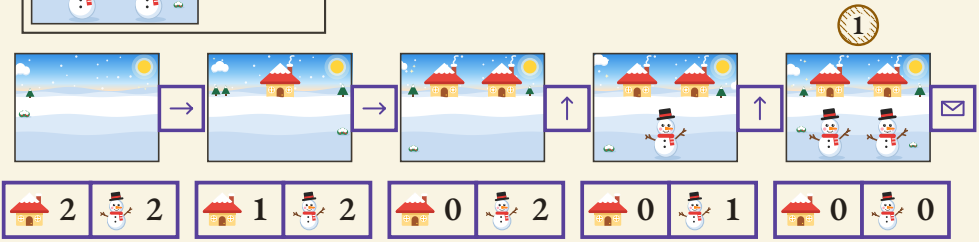


2 2	1 2	0 2	0 1

The rule finds the corner



What is missing decides the road.



Your turn again

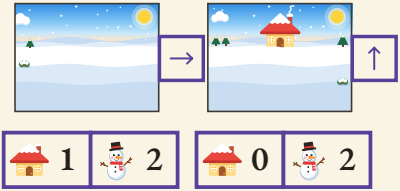






Your turn again





What is missing decides the road.

Your turn again



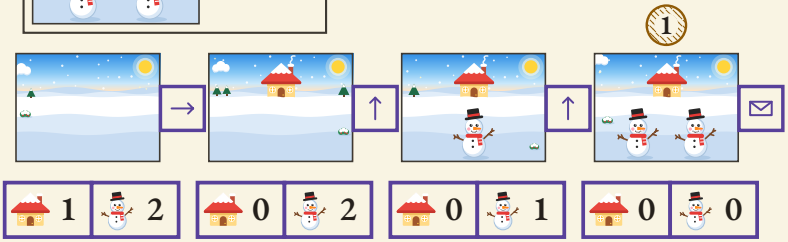
What is missing decides the road.



Your turn again



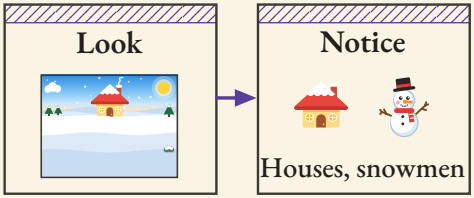
What is missing decides the road.



Look, notice, compare, step

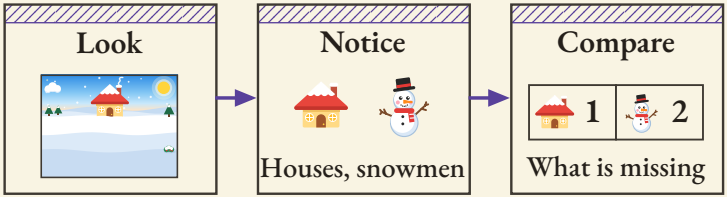


Look, notice, compare, step



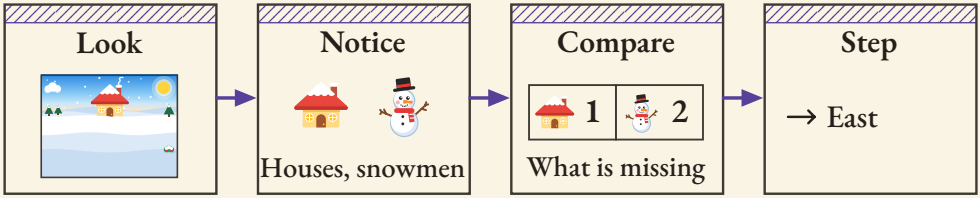
She ignores the sun, the clouds and the trees.

Look, notice, compare, step



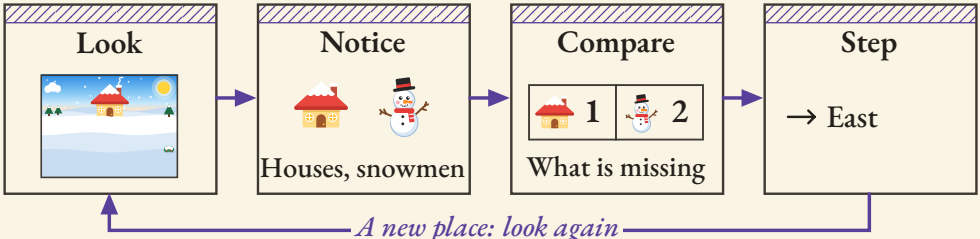
She ignores the sun, the clouds and the trees.

Look, notice, compare, step



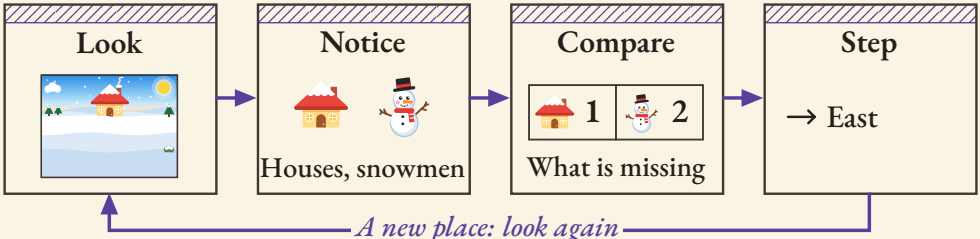
She ignores the sun, the clouds and the trees.

Look, notice, compare, step



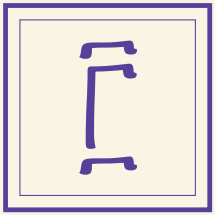
She ignores the sun, the clouds and the trees.

Look, notice, compare, step



She ignores the sun, the clouds and the trees.

Nobody told her what to notice. The coins taught her.



Section 3

Reasoning

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	Houses and snowmen	What matters in the notes

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	Houses and snowmen	What matters in the notes
Step	A road	One small step, written down

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	Houses and snowmen	What matters in the notes
Step	A road	One small step, written down
Repeat	Until she delivers	Until it has an answer

A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	Houses and snowmen	What matters in the notes
Step	A road	One small step, written down
Repeat	Until she delivers	Until it has an answer
Learn	From the coins	From checking the answers

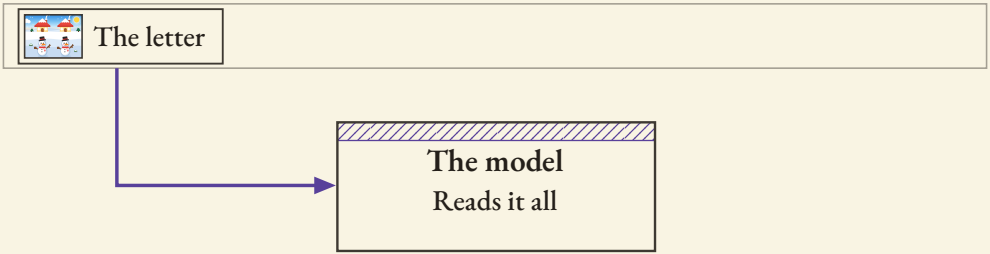
The model writes one token at a time

The conversation

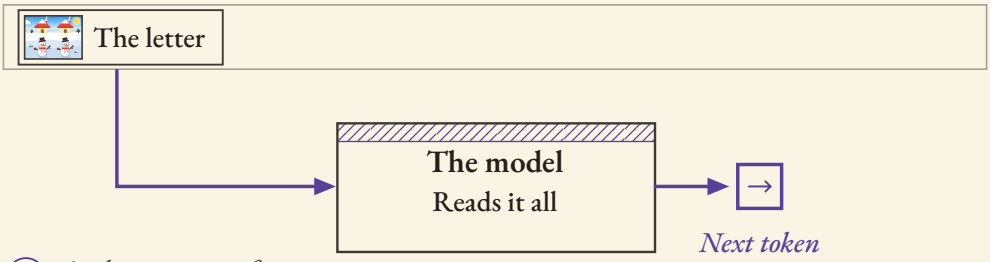
 The letter

The model writes one token at a time

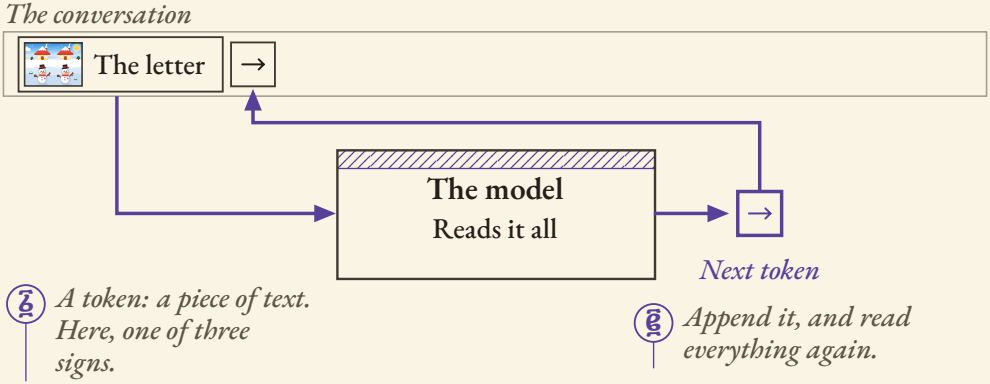
The conversation



The model writes one token at a time



The model writes one token at a time



From the conversation, four numbers

The conversation



The letter

From the conversation, four numbers

The conversation



The letter

Here, no arrow yet



0 0

From the conversation, four numbers

The conversation



Here, no arrow yet



The letter



From the conversation, four numbers

The conversation



Here, no arrow yet



The letter



The representation

Everything else in the pictures is dropped. Four numbers stay.

A difference picks the next token

Here



0	0

The letter



2	2

A difference picks the next token

Here



0 0

The letter



2 2

Letter minus here

2 2

A difference picks the next token

Here



0 0

The letter



2 2

Letter minus here

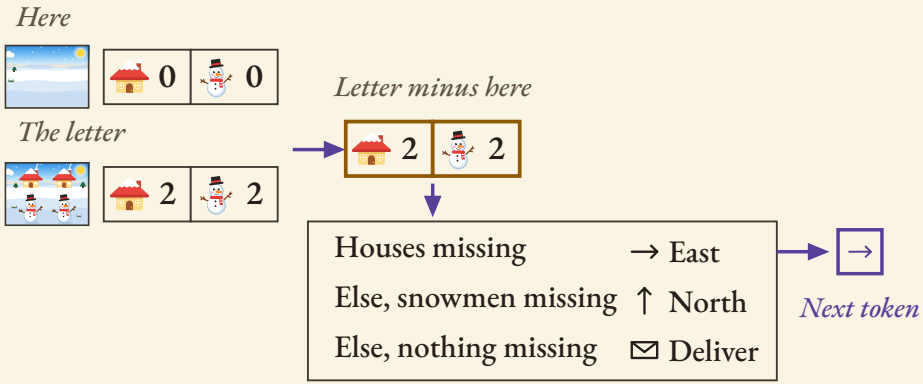
2 2

Houses missing → East

Else, snowmen missing ↑ North

Else, nothing missing ✉ Deliver

A difference picks the next token



Here is worked out from the tokens

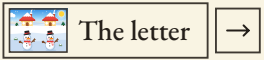

The letter

Here, no arrow yet




0 0

Here is worked out from the tokens



Here, no arrow yet



After one arrow



Each east adds a house, each north a snowman.

Here is worked out from the tokens



Here, no arrow yet *After one arrow* *After two arrows*



Each east adds a house, each north a snowman.

Here is worked out from the tokens



Here, no arrow yet



After one arrow



After two arrows



After three arrows



Each east adds a house, each north a snowman.

The model only counts the arrows.

Append, and compute again

The conversation	Here		Missing		Next
 The letter	 0	 0	 2	 2	→ East

Append, and compute again

The conversation	Here	Missing	Next
 The letter	0 0	2 2	→ East
 The letter →	1 0	1 2	→ East

Append, and compute again

The conversation	Here	Missing	Next
 The letter	0 0	2 2	→ East
 The letter →	1 0	1 2	→ East
 The letter → →	2 0	0 2	↑ North

Append, and compute again

The conversation	Here	Missing	Next
 The letter	0 0	2 2	→ East
 The letter →	1 0	1 2	→ East
 The letter → →	2 0	0 2	↑ North
 The letter → → ↑	2 1	0 1	↑ North

Append, and compute again

The conversation	Here	Missing	Next
 The letter	0 0	2 2	→ East
 The letter →	1 0	1 2	→ East
 The letter → →	2 0	0 2	↑ North
 The letter → → ↑	2 1	0 1	↑ North
 The letter → → ↑ ↑	2 2	0 0	✉ Deliver

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

Thinking

3 crates at 40 francs: 120 francs.

Each line reads the notes so far and adds one small step.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.

Each line reads the notes so far and adds one small step.

Like Mara: look, notice what matters, step.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.
10 percent of 120 is 12.

Each line reads the notes so far and adds one small step.

Like Mara: look, notice what matters, step.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.
10 percent of 120 is 12.
120 minus 12 is 108.

Each line reads the notes so far and adds one small step.

Like Mara: look, notice what matters, step.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.
10 percent of 120 is 12.
120 minus 12 is 108.

Answer: the client pays 108 francs.

Each line reads the notes so far and adds one small step.

Like Mara: look, notice what matters, step.